

Project Report: Exploratory Data Analysis on Global Climate Data

Title

Rising Temperatures: A Data-Driven Look at Climate Change Trends (1880-Present)

Date

5th May 2025

1. Introduction

Climate change is one of the most pressing issues of our time. Analyzing historical temperature records allows us to understand long-term global warming trends. This project focuses on exploratory data analysis (EDA) of global land and ocean temperature anomaly data, using Python-based data science tools.

2. Objectives

- To perform EDA on global climate data
- To uncover trends and anomalies in historical temperature changes
- To visualize global warming patterns using graphs
- To interpret the findings and discuss their relevance

3. Dataset Description

- Source: NOAA (National Oceanic and Atmospheric Administration)
- Format: CSV

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- Time Period: 1880 - Present
- Columns:
 - Date: Monthly or yearly timestamp
 - Anomaly: Temperature deviation from the average (baseline)

4. Tools & Technologies

- Programming Language: Python
- IDE: Jupyter Notebook
- Libraries:
 - pandas - data manipulation
 - numpy - numerical operations
 - matplotlib and seaborn - data visualization

5. Data Preprocessing

- Checked for missing or null values
- Parsed the date column to datetime format
- Removed irrelevant metadata rows
- Converted temperature values to float type

6. Exploratory Data Analysis (EDA)

Key Visualizations:

- Line Plot of temperature anomaly over years
- Histogram showing distribution of anomaly values
- Rolling Average Plot to show smoothed trends

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Sample Code:

```
import pandas as pd

import matplotlib.pyplot as

plt import seaborn as sns

df = pd.read_csv('climate.csv',

skiprows=4) df['Date'] =

pd.to_datetime(df['Date'])

df.set_index('Date', inplace=True)

plt.figure(figsize=(12, 6))

sns.lineplot(x=df.index, y='Anomaly',

data=df)

plt.title('Global Temperature Anomalies Over

Time') plt.ylabel('Temperature Anomaly (°C)')

plt.xlabel('Year')

plt.grid(True)

plt.show()
```

7. Key Findings

- A clear upward trend in global temperature anomalies since the late 20th century
- The 21st century contains the hottest recorded years on average
- Climate anomalies have become more frequent and extreme in recent decades

8. Conclusion

This project successfully highlights the increasing trend in global temperature anomalies, confirming the

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reality of ongoing climate change. Through visualization and statistical analysis, EDA proves to be a powerful method for interpreting climate data and informing public understanding.

9. Future Work

- Add CO2 levels and correlate with temperature
- Develop predictive models using time series forecasting
- Create an interactive dashboard with Plotly or Streamlit

10. References

- NOAA Global Climate Data: <https://www.ncei.noaa.gov>
- Our World in Data: <https://ourworldindata.org>
- Kaggle Datasets: <https://www.kaggle.com>

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